

Chapter 1 Rings

1.1 Definitions

1.2 UFD,PID,Euclidean Ring

1.3 Localization

1.4 title

1.4.1 Ring Maps of Finite Type and of Finite Representation

Definition 1.4.1.1 Let $R \rightarrow S$ be a ring map:

- $R \rightarrow S$ is of **finite type**, or that S is a **finite type R -algebra** if there exist an $n \in \mathbb{N}$ and an surjection of R -algebras

$$R[x_1, \dots, x_n] \rightarrow S$$

- $R \rightarrow S$ is of **finite presentation** if there exist integers $n, m \in \mathbb{N}$ and polynomials

$$f_1, \dots, f_m \in R[x_1, \dots, x_n]$$

and an isomorphism of R -algebras

$$R[x_1, \dots, x_n]/(f_1, \dots, f_m) \cong S$$

1.4.2 Ring Maps of Finite Type and of Finite Presentation

1.4.3 Finite Ring Maps

1.4.4 Homomorphisms Essentially of Finite Type

Definition 1.4.4.1 Let $R \rightarrow S$ be a ring map:

- $R \rightarrow S$ is **essentially of finite type** if S is the localization of an R -algebra of finite type. i.e.

$$S = M^{-1}R[x_1, \dots, x_n]$$

- $R \rightarrow S$ is **essentially of finite representation** if S is the localization of an R -algebra of finite representation

◇ **Lemma 1.4.4.2** The class of ring maps which are **essentially of finite type** is preserved under composition. Similarly for essentially finite representation.

◦ **Proof ...**

□

Under Base CHange

♠ **Proposition 1.4.4.3** ...

◦ **Proof ...**

□

Maps to Artinian Local Ring

◇ **Lemma 1.4.4.4** ...

◦ **Proof ...**

□

Essentially Finite Maps between Local Rings

◇ **Lemma 1.4.4.5** If $\varphi : R \rightarrow S$ be essentially of finite type with R and S local (but, φ may be not local) .
Then, there exists an n and a maximal ideal

$$m \subseteq R[x_1, \dots, x_n]$$

lying over m_R such that: S is a localization of a quotient $R[x_1, \dots, x_n]_m$.

◦ **Proof ...**

□

Chapter 2 Modules

2.1 Internal Hom

2.2 Tensor Product

Definition 2.2.0.1

◇ **Lemma 2.2.0.2** (Tensor Product commutes with Colimits) Let (M_i, μ_{ij}) be a system over a preordered set I , then

$$\operatorname{colim}(M_i \otimes N) \cong (\operatorname{Colim} M_i) \otimes N$$

◦ **Proof**

□

◇ **Lemma 2.2.0.3** Let $M_1 \xrightarrow{f} M_2 \xrightarrow{g} M_3 \longrightarrow 0$ be an exact sequence of R -modules and homomorphisms, and N be an R -module, then the sequence

$$M_1 \otimes N \xrightarrow{f \otimes id} M_2 \otimes N \xrightarrow{g \otimes id} M_3 \otimes N \longrightarrow 0$$

is exact. In other word, $- \otimes N$ is right exact.

◦ **Proof**

□

2.2.1 Finiteness

◇ **Lemma 2.2.1.1**

◦ **Proof**

□

2.2.2 Tensor Algebra

2.3 Limits, Colimits

2.3.1 Directed System, Directed Limits

Definition 2.3.1.1 Let $(I, \leq)$ be preordered set. Then, (M_i, μ_{ij}) of R -modules over I consists of a family of R -modules $\{M_i\}$ indexed by I , and a family of R -module maps $\{\mu_{ij} : M_i \rightarrow M_j\}_{i \leq j}$ such that for $i \leq j \leq k$

$$\mu_{ii} = id_{M_i} \quad \mu_{ik} = \mu_{jk} \circ \mu_{ij}$$

We say (M_i, μ_{ij}) is a **directed system** if I is a directed set.

◇ **Lemma 2.3.1.2** Let (M_i, μ_{ij}) be a family of R -modules over the preordered set I . The colimit of the system (M_i, μ_{ij}) is the quotient R -module

$$(\oplus_i M_i)/Q \quad Q = \langle \{\iota_i(x_i) - \iota_j(\mu_{ij}(x_i))\}_{i \leq j} \rangle_R$$

where $\iota_i : M_i \rightarrow \oplus_{i \in I} M_i$ is a natural inclusion.

We denote the colimit

$$\operatorname{colimit}(M) = \operatorname{colim}_i M_i$$

Denote $\pi : \oplus_{i \in I} M_i \rightarrow M$ the projection map and $\phi_i = \pi \circ \iota_i : M_i \rightarrow M$.

◦ **Proof ...**

□

◇ **Lemma 2.3.1.3** Let (M_i, μ_{ij}) be a system of R -modules over the preordered set I .

Assume that I is directed. The colimit of the system (M_i, μ_{ij}) is canonically isomorphic to the module M defined as follows:

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◦ **Proof ...**

□

◇ **Lemma 2.3.1.4** Let (M_i, μ_{ij}) be a directed system. Let $M = \operatorname{colim} M_i$ with

$$\mu_i : M_i \rightarrow M$$

Then, $\mu_i(x_i) = 0, x_i \in M \Leftrightarrow$ There exists $j \geq i$ such that $\mu_{ij}(x_i) = 0$

◦ **Proof ...**

□

Morphisms of Directed Systems

◇ **Lemma 2.3.1.5** ...

◦ **Proof ...**

□

◇ **Lemma 2.3.1.6** ...

◦ **Proof ...**

□

2.3.2 Directed Limit in Index Category

♠ **Proposition 2.3.2.1** ...

◦ **Proof ...**

□

2.4 Characterizing Finite and Finitely Presented Modules

2.5 Finite Case of Modules and Maps

2.5.1 Finite Modules, and Finitely presented modules

Definition 2.5.1.1 Let R be a ring and M is R -module:

- We say M is a **finite** R -module, or finitely generated module if there exists $n \in \mathbb{N}$ and $x_i \in M$, with exact sequence

$$\oplus Rx_i \rightarrow M \rightarrow 0 \quad \text{or} \quad R^{\oplus n} \rightarrow M \rightarrow 0$$

- M is **finitely represented** if there exists $m, n \in \mathbb{N}$ and exact sequence

$$R^{\oplus m} \rightarrow R^{\oplus n} \rightarrow M \rightarrow 0$$

which implies:

- M is finitely generated;
- The relation between the generators are also finitely generated.

such choice of exact sequence is called a **presentation**.

◇ **Lemma 2.5.1.2** Let R be a ring, and

$$\alpha : R^{\oplus n} \rightarrow M \quad \beta : N \rightarrow M$$

be R -module maps. If $\text{Im}(\alpha) \subseteq \text{Im}(\beta)$, then there exists an R -module map $\gamma : R^{\oplus n} \rightarrow N$ such that $\alpha = \beta \circ \gamma$.

$$\begin{array}{ccc} N & \xrightarrow{\beta} & M \\ \uparrow \gamma & \nearrow \alpha & \\ R^{\oplus n} & & \end{array}$$

◦ **Proof** Let $\alpha(e_i) = \beta(n_i)$. Then, consider the mapping

$$\gamma : R^{\oplus n} \rightarrow N \quad (a_1, \dots, a_n) \mapsto \sum a_i n_i$$

□

◇ **Lemma 2.5.1.3** Let R be a ring, and $0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$ be a short exact sequence of R -modules:

- If M_1 and M_3 are finite R -modules, then M_2 is a finite R -module;
- If M_1 and M_3 are finitely presented R -modules, then M_2 is finitely presented;
- If M_2 is a finite R -module, then M_3 is a finite R -module;
- If M_2 is finitely presented R -module and M_1 is a finite R -module, then M_3 is a finitely represented R -module;
- If M_3 is finitely presented R -module and M_2 is a finite R -module, then M_1 is a finite R -module.

◦ **Proof**

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□

◇ **Lemma 2.5.1.4** Let R be a ring and let M be a finite R -module. There exists a filtration by finite R -module

$$0 \subseteq M_0 \subseteq M_1 \subseteq \cdots \subseteq M_n = M$$

such that each quotient M_i/M_{i-1} is isomorphic to R/I_i for some ideal I_i of R .

◦ **Proof ...**

□

◇ **Lemma 2.5.1.5** Let $R \rightarrow S$ be a map, and M be an S -module. If M is finite as an R -module, then M is finite as an S -module.

◦ **Proof ...**

□

2.6 Base Change

Chapter 3 Associative Algebra

Chapter 4 Polynomial Algebra

4.1 Discriminant

Chapter 5 Matrix Algebra